

## **Unnamed\_Unit**

### **Unit 3 - Building a Simple Restroom Indicator**

#### **PDF Documents**

##### **PDF Document 1: 1737528865209-Chapter 5 - Unit 3 - Building a Simple Restroom Indicator.pdf**

This PDF document is included in the unit materials.



## UNIT 3 BUILDING A SIMPLE RESTROOM INDICATOR

In this section, we'll create a simple restroom indicator that helps show if a restroom is occupied or available. We will use a dial to adjust the turn value and LEDs to indicate whether the restroom is in use.

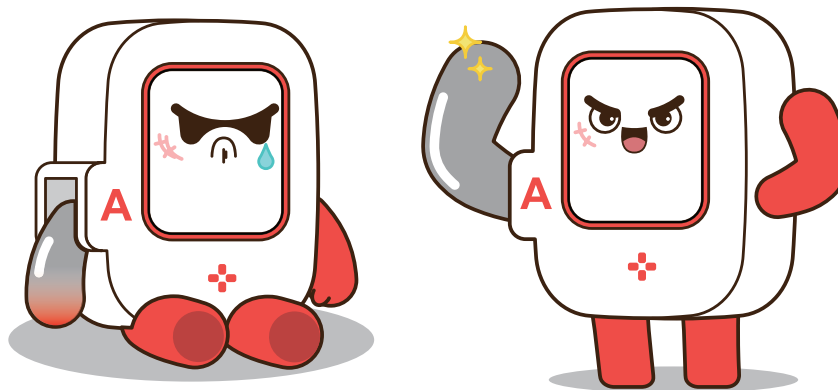
### UNDERSTANDING THE RESTROOM INDICATOR

Our restroom indicator uses a dial and two different colored LEDs (red and green) to show the status of the restroom:

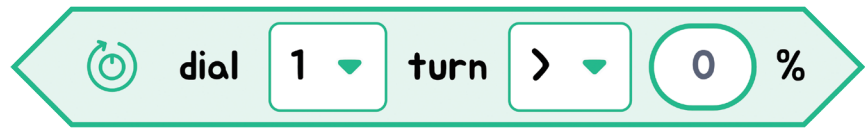
**DIAL ADJUSTMENT** The dial helps us set the turn value, which tells us if the restroom is occupied. The turn value is measured in percentage.

**LED LIGHTS** We use two LED lights to show the status of the restroom:

- **Red LED** Lights up when the turn value is 50% or more, indicating that the restroom is occupied.
- **Green LED** Lights up when the turn value is 50% or less, indicating that the restroom is available.

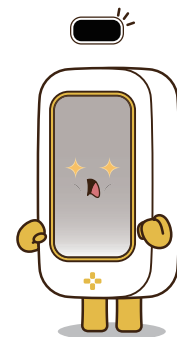
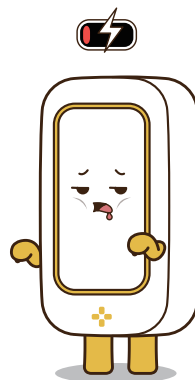
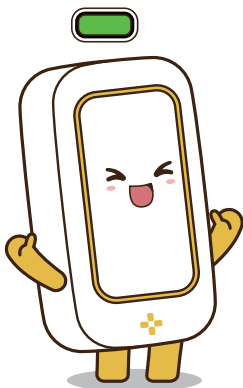


Let's look detail about the **Dial** module.



In Dial, there is 4 values.

| 1. Turn(%)                                                                       | 2. Angle(°)                                                                    |
|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
|                                                                                  |                                                                                |
| <p>You can change the position by turning the knob</p> <p>Range: 0 % ~ 100 %</p> | <p>You can change the angle by turning the knob.</p> <p>Range: 0 ° ~ 360 °</p> |





## Activity: Create Restroom Indicator

### 1 Materials Needed:



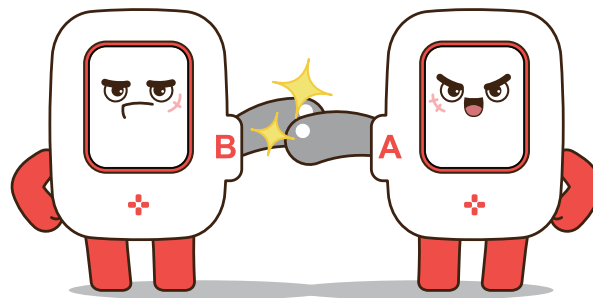
Network



Dial



LED



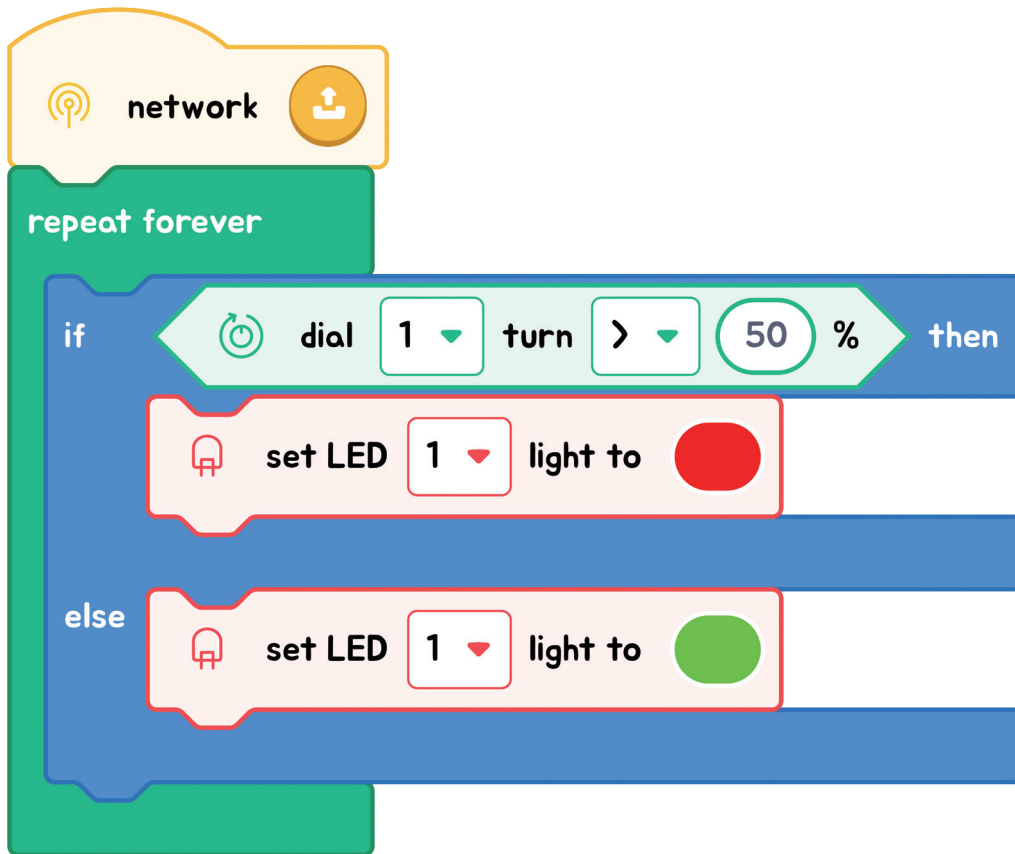
### 2 Code story

If Dia turn is over than **50%**

↓ true  
Set LED as **RED**

↓ false  
Set LED as **GREEN**

## Code Quest



### Checkpoint

If we change the turn what will happen?

| 50 → 30 | 50 → 70 |
|---------|---------|
|         |         |